

Numerical PDE Quiz 4

June 2, 2026

Consider the one-wave equation $u_t + au_x = 0$ with constant wave speed $a \neq 0$ subject to periodic boundary condition.

1. Use the interpolation approach to derive the first order upwind method. (Assume $a < 0$).
2. (a) Briefly state the definition and core idea of the CFL condition.
(b) Explain why the CFL condition is a necessary condition for the convergence.
(c) Give one example of a numerical method that satisfies the CFL condition but is not convergent.
3. (a) Define the courant number as $r = \frac{|a|\tau}{h}$. Sketch the numerical domain of the upwind method and state the range of r for which each method satisfies the CFL condition.
(b) Show the $\max_j u_j^{n+1} \leq \max_j u_j^n$ for the upwind method with about CFL constraint.
4. Apply the conserved Lax-Wendroff method to the inviscid Burgers' equation $u_t + \left(\frac{u^2}{2}\right)_x = 0$:
(a) Derive the complete finite difference scheme.
(b) Explain what "conservative form" means for this scheme.